WHAT AN UPSTREAM SOUND-EVENT TAG COSTS, AND WHICH PART OF IT: CLAIM, PACKAGING, AND DECLARED CONFIDENCE

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ABSTRACT

An upstream sound-event tagger's decision usually enters a multimodal model's prompt as a sentence. It carries three things at once: the claim, its packaging, and the declared confidence. Prior work leaves them unseparated. We separate them, holding the signal and the claim wording fixed, across six models, then through a real CLAP cascade. The claim does most of the damage: forwarding the detector's decision costs accuracy in every cascade cell that survives an answer-order audit, and not reliably in the fifth. Naming the field Confidence rather than Priority moves the model two to four times further toward the claim, and the swing survives where a shift common to true and false claims would leave none; on accuracy it is smaller, clearing zero in half the order-robust cells. Packaging governs how much lands. Every admitted reader beats the detector, so we locate where forwarding starts to pay: a per-model break-even agrees with the measured crossing.

***Index Terms*—** Sound Event Detection, Decision Fusion, Audio–Visual Language Models, Evidence Conflict, Cascaded Systems

1. INTRODUCTION

A practical audio–visual system rarely consists of one model. A sound-event tagger runs first, its output is written into a prompt [1], and a multimodal model reads that text together with the audio and video. That interface is plain language, and it carries not only a claim but an assertion of how strongly the upstream stage believes it.

Studies of this failure mode vary the claim and its framing together, measuring a confident assertion as a whole. A designer, though, chooses whether to forward the tag at all and, if so, how to serialise the claim and its confidence. Those are separate levers, and answering which one to pull requires moving them one at a time.

Our contributions are:

- A decomposition of the upstream message into claim, format and declared confidence, each held fixed against the others and reported at the grade its design supports: inserting the claim and replacing the field name are single-factor, while the format is a bundle of source, framing and length, so its damping is measured rather than attributed. The setup rungs are descriptive. All on a held-out set never used for exploration.
- A test of the claim-aligned contrast against a restricted common-shift account, which cancels in β by construction and predicts no swing on items the model answers incorrectly unaided; the swing persists there. The symmetric term reverses sign, as shrinkage predicts, and is reported as unidentified.
- A measurement of what each part costs in a real cascade: which one dominates, which one is reliable only on the margin, and which one governs how much of either lands. We report a per-model break-even that an error-repair sweep tracks, with the eligibility flow and a backbone audit.

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2. RELATION TO PRIOR WORK

Evidence–text conflict appears as sycophancy in text-only models, as image–text conflict in vision–language models [2], and in audio–visual models as hallucination benchmarks [3–5], decoding-time corrections [6–8] and counterfactual audits of whether acoustic evidence is used at all [9–14].

To our knowledge no prior study separates the declared confidence from a matched non-epistemic field on a byte-identical claim, or prices the result against a real detector's operating accuracy. Video-LLM sycophancy benchmarks vary the misleading text as a whole [15, 16]; authority-bias work grades the endorser [17, 18] — our Source factor; [19] fixes the claim and intervenes on the video. In audio, SYAUDIO [20] grades assertion strength by rewording the whole assertion and DEAF [21] grades levels of textual override with the framing fixed; in text, epistemic markers already shift accuracy [22], [23] crosses a suggestion's correctness with its stated sureness, [24] separates conformity from uncertainty, and [25, 26] calibrate a model's own verbalised uncertainty — ours is the upstream stage's. We do both, on a logit margin and on a real cascade.

3. PROTOCOL

Attributes. Sound event: 12 VGGSound [27] classes; negatives ask about another of the 12, keeping distractors comparable in kind. Colour asks whether the clip is greyscale, noise whether a steady hiss covers it, silence whether a 1.5 s mid-clip gap is muted; all three are constructed. CLAP's calibrated $P(\text{present})$ separates 9 of 12 classes by 0.26–0.56, but only 0.11–0.24 for the 3 eating and sneezing ones. The declared-confidence ladder is monotone in all 5 sound cells with those classes in or out; nothing rests on the exclusion.

The message and its controls. Metadata precedes the question, verbatim as `Source: anonymous system / Claim: ``The clip does not contain the sound of X.'' / Confidence: high`, followed by `Does this clip contain the sound of X? (A) Yes (B) No.`

condition	Source	Claim	third field
signal only	`--`	`--`	`--`
empty block	`---`	`""`	`Confidence: ---`
source-only	`anon`	`""`	`Priority:` c
non-epistemic	`anon`	`"F"`	`Priority:` c
declared conf.	`anon`	`"F"`	`Confidence:` c

The two contrasts the paper rests on are single-factor: source-only and non-epistemic differ in the claim alone, non-epistemic and declared confidence in the field's name alone. The setup rungs above them are not, and we do not read them causally. The ladder runs very low to very high, with a numeric scale (0.15/0.35/0.60/0.85/0.95) as a second, and the claim sentence is byte-identical across every claim-bearing condition.

Perceptual-eligibility gate. Fixed before any metadata condition was scored, and evaluated on the no-metadata pass only — no condition containing metadata is read before a cell is admitted. A cell is admitted when both-direction recall ≥ 0.60 and unaided accuracy $A_0 \geq 0.60$; below that the model cannot judge the property; failing recall means "correct" is position. A cell is one (model, attribute) pair. Cells with $A_0 > 0.95$ are ceiling: accuracy has no headroom there, the margin does. Of 53 cells run, 32 floor, 7 fail recall, 5 are ceiling and 9 eligible; those 9+5 admissible ones plus 2 kept for the position audit were rerun in the swapped order, and 10 passed both (order-robust) — 10 headline, 7 ladder, 6 field-name and 4 cascade cells, the narrower sets following from what each cell was run on, not what it showed; 5 of the 10 headline cells hit ceiling in at least one order. No cell is selected out of the headline contrast. Two cases are reported separately: Audio Flamingo 3/sound carries the contrast but fails the swapped gate, and silence, which on Qwen2.5 and Qwen3 passes the first order and floors under the swap.

Scoring. One forward pass; the answer is the larger of the two option-letter logits, so decoding is deterministic; free-form generation is not tested. We study Qwen2.5-Omni-7B [28], Qwen3-Omni-30B-A3B [29], VideoLLaMA2.1-7B-AV [30], Qwen2-Audio-7B [31] and Audio Flamingo 3 [32]. We add Nemotron 3 Nano Omni (BF16, reasoning disabled) [33], whose Mamba2-Transformer LLM, Parakeet speech encoder and C-RADIO vision tower differ from all five. Phi-4-multimodal [34] was screened and excluded for answering by option position: on the same 400 clips it is right 99.5% when the answer is (A) and 1.5% when it is (B).

Metrics. POR is the probability of following the false claim among items answered correctly unaided, macro-averaged over the two claim directions. The truth-aligned margin is $m = \log p(\text{correct}) - \log p(\text{incorrect})$, the log-likelihood ratio between the two answers. Writing Δ_{true} and Δ_{false} for the margin change from the signal-only prompt under a true and a false claim at level c, we report the symmetric and claim-aligned contrasts

$$\alpha_c = \tfrac{1}{2}(\Delta_{\text{true}} + \Delta_{\text{false}}), \qquad \beta_c = \tfrac{1}{2}(\Delta_{\text{true}} - \Delta_{\text{false}}).$$

This is an exact re-parameterisation of two measurements into two numbers, not a testable model, and we do not read the two terms as separate mechanisms. Intervals are bootstrapped over source videos; paired conditions use an exact McNemar test with Holm correction over the K=18 prespecified per-order confidence contrasts and, separately, the 4 bare-cascade cells; comparisons outside those two families are descriptive.

4. WHAT EACH PART OF THE MESSAGE COSTS

Inserting the claim is the largest step in Table 1's walk from the signal-only prompt to a confident false claim (−1.1 to −4.0 log-odds), over twice the next in 6 of the 7 rows; the declared confidence adds −0.1 to −1.0 at high, 5–44% of it.

At low the same increment is positive in all 7 ladder cells (+0.08 to +0.97): weak confidence pushes the answer back toward the signal.

The non-epistemic field also carries a true claim, so the claim-aligned contrast is defined for it too. Between high and low it swings

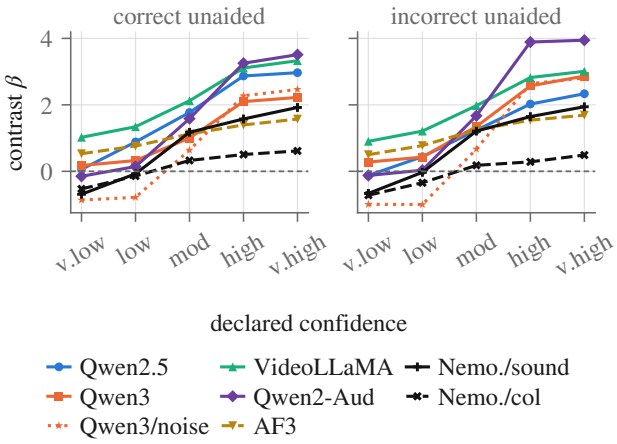


Fig. 1. β against declared confidence, on unaided-correct items (left) and their complement (right); curves use the first order, with left-panel n of 308–382 per model; the right panel holds the $400-n$ complement (ceiling cells: 19/18). A common-shift account predicts no swing in the right panel; the swing persists.

by 0.65–3.02 log-odds under Confidence but only 0.20–0.80 under Priority. Paired on the same items and averaged over both answer orders, the difference is 0.46–2.22 log-odds over all 10 headline cells, every bootstrap interval excluding zero, a ratio of 2.1–3.9× that stays tightly bounded despite a five-fold spread in absolute size.

Epistemic content, not a graded word. If the swing came from the words high/low rather than from what the field asserts, a non-epistemic field with the same words would reproduce it. It does not: the right half of Table 1 sets the field names side by side. The non-epistemic names (Priority, Urgency, Metric-X) recover 5–84% of the Confidence contrast, under half in 14 of the 21 cell–field pairs. A field could still matter by carrying meaning alone; Prior tests that: epistemic vocabulary, same block and graded value, but about the class base rate, not this clip. It reaches 25% of the Confidence contrast (median over 10 cells; Table 1 lists seven), is smaller in all 10 and nearer the non-epistemic side in 9. Nor is it acting where β cancels: a base-rate field should push toward presence on both branches, but its presence-aligned shift does so in only 5 of 10 cells (mean 0.19 against Confidence's 0.54). The content has to be about the claim.

5. THE CLAIM-ALIGNED CONTRAST, AND WHAT IS NOT IDENTIFIED

β rises monotonically with declared confidence in every one of the 7 ladder cells and in Audio Flamingo 3, reaching +0.61 to +3.51 at the top rung. On Nemotron, Qwen2-Audio and Qwen3/noise the ladder starts below zero, so a claim declared at very low is not merely discounted but read as evidence against itself. The response is also damped: on the numeric scale, where the value is explicit, β is close to linear in $\text{logit}(c)$ ($R^2 \geq 0.86$) with slope 0.25–0.96, where a calibrated independent sensor taken at face value would give one.

The test β has to pass. Conditioning on unaided-correct items makes their margin positive by construction, so output shrinkage mimics the same signs. The null we test is restricted: a confidence term common to both branches cancels exactly in β and predicts no swing on the complement. The swing persists under the prespecified first order (Fig. 1, right) and, in the two cells rerun in the swapped order (Qwen2-Audio, Nemotron/sound), there too: the complement's

Table 1. Margin contrasts (log-odds) at high on unaided-correct items, order-averaged. Left, each rung's step from the one above (descriptive). Right, a different quantity — each field's own $m_{\text{high}} - m_{\text{low}}$ swing on the false-claim block — so the halves are not subtractable. ^{s}: sensitivity.

	one-step contrasts				epistemic				non-epistemic		
model	block	+src	+claim	+conf.	Conf.	Reliab.	Certain.	Prior	Priority	Urgency	Metric-X
Qwen2.5-Omni	+0.59	−0.47	−3.39	−0.53	−2.33	−2.11	−2.18	−1.36	−0.85	−0.73	−0.92
Qwen3-Omni	−1.22	−0.17	−1.63	−0.72	−1.35	−0.61	−1.74	−0.34	−0.52	−0.35	−0.78
VideoLLaMA2.1	−1.01	−0.06	−4.01	−0.19	−1.03	−1.83	−2.09	−0.24	−0.64	−0.58	−0.70
Qwen2-Audio	−0.49	+0.27	−3.10	−0.97	−2.00	−1.88	−2.55	−0.38	−0.17	−0.09	−0.42
Audio Flamingo 3^{s}	−0.51	+0.14	−1.14	−0.09	−0.54	−0.22	−0.19	−0.15	−0.23	−0.07	−0.08
Nemotron/sound	−0.23	+0.02	−2.18	−0.22	−1.68	−1.75	−1.54	−0.33	−0.58	−0.38	−1.01
Nemotron/colour	+0.21	+0.02	−1.17	−0.11	−0.45	−0.49	−0.46	+0.00	−0.26	−0.22	−0.38

Table 2. Override rate (macro-POR, %) under high vs. low declared confidence. $|C|$ counts items answered correctly unaided, per order; spread is the across-order range at high/at low. p is the exact McNemar of the worse order, raw; Holm over the $K{=}18$ prespecified per-order comparisons rejects 17.

model	cell	$\lvert C\rvert$	high	low	spread	p
Qwen2.5$^{c\text{-}1}$	sound	381/368	49.1	22.1	0.1/5.1	1.6e-27
Qwen3$^{c\text{-}1}$	sound	382/362	32.3	10.3	6.4/7.2	4.2e-22
Qwen3	noise	346/346	52.1	10.9	15.8/15.8	4.1e-25
VideoLLaMA2.1	colour	324/368	93.9	60.2	2.8/6.5	3.1e-33
Qwen2-Aud	sound	308/344	59.4	24.7	7.1/30.1	1.7e-21
Nemotron	sound	368/364	55.7	22.9	6.4/10.9	1.1e-32
Nemotron$^{c\text{-}2}$	colour	349/384	7.2	1.5	6.3/2.4	1.2e-04
fails admission in one order — sensitivity only						
Audio Flamingo 3	sound	346/276	50.5	49.6	2.5/0.8	1.0e+00
Qwen2.5	silence	294/233	55.4	50.0	5.0/0.0	9.8e-04

$N{=}400$/order; $A_0{=}|C|/400$. ^{c}ceiling > 0.95 (-1/-2: first/swapped order). Cells without POR headroom are not rows.

top-minus-bottom swing is $+4.37$ and $+3.75$ log-odds, both intervals above zero and the latter larger than in the first order. Rescaling common to both branches is not excluded by this test; the Priority control shows that a generic graded-value effect is insufficient.

The symmetric term α fails the same test — refit on unaided-incorrect items it reverses sign in 6 of 7 ladder cells, as shrinkage predicts — so we report it as observed, not identified.

6. ABLATIONS

Withholding the clip raises the contrast in 3 of 4 masked cells, and outside the gate β runs 1.86–5.30, above the admitted maximum at high (3.40) in 5 of 6 floor cells: alignment is strongest where perception is weakest. A reworded question keeps the POR gap on the two Qwen sound cells (18.0–24.0 pp).

Which cells these are. The audio claim rests on 4 sound-event cells (Table 3); the other 4 colour and 2 noise cells make the field effect a property of the interface, not of one attribute.

Answer-option order. Inverting the options keeps sign and one-sidedness but not magnitude — Qwen3/noise's largest gap moves $57.0 \to 25.3$ — so all both-order numbers are order-averaged. VideoLLaMA2.1's unaided sound accuracy falls to chance (54.5) under the swap, so those cells fail the gate there and are not reported.

Table 3. The 10 headline cells and the Audio Flamingo 3 sensitivity row. A_0 is unaided accuracy, order-averaged; $\beta_{\text{C}}/\beta_{\text{P}}$ are the high-minus-low swings of β under Confidence and Priority (log-odds), Δ their order-averaged paired difference with a source-clustered 95% interval (393–399 videos, 2000 replicates; within-order per-item s.d. 0.16–1.26). ^{c}: an order exceeds 0.95. ^{s}: sensitivity. g4: strongest graded-noise level.

model	attribute	A_0	β_{C}	β_{P}	Δ [95% CI]
Qwen2.5-Omni	sound	93.6^{c}	2.16	0.79	1.37 [1.26, 1.49]
Qwen3-Omni	sound	93.0^{c}	1.77	0.64	1.13 [1.09, 1.18]
Qwen3-Omni	noise	86.5	2.83	0.76	2.07 [2.02, 2.13]
VideoLLaMA2.1	colour	86.5	1.26	0.59	0.67 [0.66, 0.69]
Qwen2-Audio	sound	81.5	3.02	0.80	2.22 [2.13, 2.31]
Audio Flamingo 3^{s}	sound	77.8	0.66	0.24	0.42 [0.40, 0.44]
Nemotron	sound	91.5	1.91	0.50	1.40 [1.36, 1.45]
Nemotron	colour	91.6^{c}	0.65	0.20	0.46 [0.42, 0.49]
Qwen2.5-Omni	colour	99.9^{c}	0.90	0.23	0.67 [0.63, 0.72]
Qwen2.5-Omni	noise (g4)	83.9	1.56	0.54	1.02 [0.95, 1.09]
Qwen3-Omni	colour	99.6^{c}	0.95	0.38	0.56 [0.53, 0.60]

7. THE DECOMPOSITION ON A REAL CASCADE

CLAP (clap-htsat-unfused) scores each clip against the query class verbatim, unlike fixed-vocabulary taggers [36]. Its threshold and band precisions are cross-fitted over a five-fold split of the source videos, and everything below is pooled over the five held-out folds, reaching 74.8% (per-fold s.d. 4.1).

What the detector declares needs defining. CLAP's score measures similarity to the positive phrasing and its threshold is not at 0.5, so $\max(s, 1-s)$ is confidence about presence, not about the decision — on a third of clips the opposite of the claim sent. We fit $p = P(y{=}1 \mid s)$ by isotonic regression and declare $q = p$ or $1-p$ by the decision; the bands (training-fold quintiles of q) then rise out-of-fold (66.7–92.3%), as do the second checkpoint's. All 5 cascade cells are sound events.

Forwarding loses, and the branch arithmetic says why. A true claim repairs only items already wrong, so its gain is capped by $1-A_0$; a false one is capped by A_0, and mixing at claim correctness π gives $\pi^{\star} = -G_{\text{false}}/(G_{\text{true}} - G_{\text{false}})$. On branches matching the claim-only policy a true claim already costs 1.3–3.6 points in 2 of 5 cells, leaving no $\pi^{\star}$. All 5 cascade cells are predicted to lose; the 4 order-robust rows do, by 4.2–9.8 points, and Audio Flamingo 3 most of all (Table 4). Mixing linearly assumes claim errors independent of item difficulty; they are not, so $\pi^{\star}$ is approximate; the measured column is what the recommendation rests on. The bare column sends the detector's actual decisions: it costs 6.6–11.1 points in every order-robust cell, and 5.1–10.8 on the second check-

Table 4. A real cascade, order-averaged. CLAP [35] tags each clip and its decision becomes the claim. $\pi^\star$ is the in-block break-even. Brackets are source-clustered 95% intervals; the in-block ones (not shown) also exclude zero. ^{s}: sensitivity.

		forward claim (Δ omit, pp)			field (Δ block, pp)		
model	omit (%)	bare		in block	verbal	numeric	$\pi^\star$
Qwen2.5-Omni	93.6	-9.8	$[-12.9, -6.7]$	-9.8	-0.2	$+0.9$	> 1
Qwen3-Omni	93.0	-9.5	$[-12.5, -6.6]$	-5.1	-1.6	-0.4	0.97
Qwen2-Audio	81.5	-6.6	$[-11.2, -1.7]$	-4.2	-2.9	$+0.0$	0.76
Audio Flamingo 3^s	77.8	-1.1	$[-5.1, 2.9]$	-10.2	-0.5	$+1.9$	> 1
Nemotron/sound	91.5	-11.1	$[-15.2, -6.9]$	-5.6	-7.1	-3.0	0.87

point (larger_clap_general), whose per-cell intervals all exclude zero. Qwen2-Audio as detector (its no-metadata answer, a different family at 77.8% out-of-fold) costs 5.5–9.0 points in 3 of 3 cells: the loss is not specific to CLAP. Holm over the 4 bare-cascade cells leaves all below 0.05 (largest adjusted $p = 5e - 04$). It is also direction-asymmetric (post hoc): of the detector's 55 false absences and 46 false presences, absence carries 69–91% of the false-claim loss.

The break-even predicts the crossover, and a stronger detector crosses it. One seeded subset of the detector's errors is corrected per fold (the rest are its own); the endpoints are the real run and the wholly true claim. The loss rises monotonically and crosses zero 0.8–2.0 pp above the predicted bare $\pi^\star$, a paired gap whose interval covers zero in all 4 cells (Table 5). Two further repair seeds put the crossing in the same grid interval in 7 of 8 cell–seed pairs. A stronger probe (an L_2 logistic on both checkpoints' embeddings and similarities, cross-fitted on the same source folds; 0.897 out-of-fold) lands on the same line: $+8.4$ on Qwen2-Audio ($[+4.2, +12.3]$), $-1.6/-1.7$ where $\pi^\star$ exceeds it, -0.4 $[-3.7, +3.1]$ at Nemotron's knife-edge, where the preregistered sign was positive — each within 0.5 pp of the sweep's π=0.90 column. Qwen2-Audio's cascade then reads 89.9% against the probe's own 89.7%: the gain is over the reader, not the detector. Score-level fusion [37] of the reader's margin and the detector's decision gains at most $+2.1$ points, on the weakest reader only, and causes no comparable degradation.

Packaging damps the claim rather than reversing it. The claim moves the model less inside a Source/Claim block than as the same bare sentence. The gap between a true and a false claim shrinks to 36–79% of the bare span (median 61%) in all 4 cascade cells, with every interval below 100%; Audio Flamingo 3 gives 34%. Both directions shrink; on Qwen2.5 the span contracts while both policies cost the same. The damping is the block's: a true claim moved into it costs 2.8–6.5 points in all 4 cascade cells, of which Confidence returns 1.3–3.9, clearing zero in 3. On colour it is 12–24%: the damping recurs beyond audio. The exception is Audio Flamingo 3, which fails the order gate: there the block costs 9.1 points more than the bare sentence, so the recommendation is for order-robust readers. The numeric field damps too: written as the band's training-fold precision at the same declared level, it compresses the response to 52–64% of the verbal slope (median 63%) in all 4 cells, and costs 1.1–4.1 points less than the word in every cell of both detectors. Against a block with no field it gains in 1 of 4 cells and loses in 2.

A prompt caveat is not the lever. Appending "`This tag may be wrong; judge from the recording itself.`" to the bare sentence helps in 1 of 4 cells, hurts in another, and all 4 still trail omitting, while the block matches or beats the bare sentence in all 4 cells ($+0.0$ to $+5.5$); an oracle confidence gate, its threshold swept on the test set, gains ≤ 1.25 points (descriptive).

What the field moves. On the margin the declared field costs a fraction of the claim's step and is the most reliable: its swing beats the control in every cell. On accuracy it costs 0.2–7.1 points, clears zero in 2 of 4 cascade cells, and on Nemotron/sound exceeds the in-block claim (7.1 against 5.6). The field is where the effect shows; the format is where the leverage is.

Table 5. Error-repair sweep: accuracy change of bare forwarding vs. omitting at each target π. $\pi^\star$ is the bare-sentence break-even (Table 4's is in-block); its source-clustered 95% half-width is 0.03; crossings are interpolated between columns.

model	.748	.80	.85	.90	.95	1.0	$\pi^\star$
Qwen2.5-Omni	-9.8	-6.9	-3.9	-1.5	$+1.9$	$+5.3$	0.90
Qwen3-Omni	-9.5	-6.6	-3.9	-1.4	$+0.8$	$+3.8$	0.92
Qwen2-Audio	-6.6	-1.4	$+3.4$	$+8.4$	$+13.6$	$+18.5$	0.81
Nemotron	-11.1	-7.5	-3.5	$+0.1$	$+3.9$	$+8.1$	0.89

8. LIMITATIONS

All six readers are open and $\leq$30B; the first five share a Qwen text backbone, and Nemotron shares none of the three components yet reproduces the contrast. All cells come from a single corpus, and every admitted one reads more accurately than CLAP. The rule admits from $A_0 \geq 0.60$ but admitted cells start at 81.5, so the cost is measured where the reader is strong and the break-even highest; the cells below it failed order-robustness (Qwen2.5-Omni/silence: 73.5 in the order it passed). The stronger probe inverts that ranking on one reader, where forwarding wins; whether that holds elsewhere is untested. A blind pass over 98 clips of the 3 worst-separated classes agrees with VGGSound on 94.7% of the 95 it judged; dropping the 5 disagreements and 3 unsure clips leaves the bare loss at 5.4–10.2. Relabelling the field Reader or Detector confidence shrinks the swing by 0.2–1.0 log-odds but it still beats Priority in all 4 cells.

9. CONCLUSION

Break-evens depend on how the tag is written. Every bare-sentence threshold we measured exceeds this detector's accuracy, and the in-block ones differ from those, so clearing a threshold under one serialisation does not license another. Against this detector, omitting beat every text-forwarding policy we tested. A probe that outranks the reader reverses that on one cell, so the choice turns on the two stages' relative accuracy and on where their errors coincide. When a claim must go, the block is the lever, not a caveat; a numeric field costs less than the verbal band but is not consistently better than no field.

10. COMPLIANCE WITH ETHICAL STANDARDS

This work used the public VGGSound corpus and public pre-trained models. No studies with human or animal subjects were performed by any of the authors, so no ethical approval was required. The authors declare no conflict of interest.

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